



Web-Based Application Development (CSE3023)

Lab Report 6

Group:

K1

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Program:

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Lab Exercise

1. Create a table known as `userprofile` using database schema `CSE3023` using these attributes.
 - `username` as a character length 15 and must be primary key
 - `password` as a character length 10
 - `firstname` as `varchar(50)`
 - `lastname` as `varchar(50)`

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 username 	char(15)	utf8mb4_0900_ai_ci		No	None			 Change  Drop  More
☐	2 password	char(10)	utf8mb4_0900_ai_ci		No	None			 Change  Drop  More
☐	3 firstname	varchar(50)	utf8mb4_0900_ai_ci		Yes	NULL			 Change  Drop  More
☐	4 lastname	varchar(50)	utf8mb4_0900_ai_ci		Yes	NULL			 Change  Drop  More

2. Create insertUser.html as the main interface to register a new user.

```
6 <html>
7 <head>
8   <title>TODO supply a title</title>
9   <meta charset="UTF-8">
10  <meta name="viewport" content="width=device-width, initial-scale=1.0">
11 </head>
12 <body>
13   <h1>Register User</h1>
14
15   <div>
16     <form action="processUser.jsp" method="POST">
17       <label for="username">Username: </label>
18       <input name="Username" id="username" required type="text"/>
19       <br>
20
21       <label for="password">Password: </label>
22       <input name="Password" id="password" required type="password"/>
23       <br>
24
25       <label for="firstName">First Name: </label>
26       <input name="FirstName" id="firstName" required type="text"/>
27       <br>
28
29       <label for="lastName">Last Name: </label>
30       <input name="LastName" id="lastName" required type="text"/>
31       <br>
32
33       <p>
34         <input type="submit" value="Register User" id="btn"/>
35       </p>
36     </form>
37   </div>
38 </body>
39 </html>
```

Register User

Username

Password

First Name

Last Name

3. Create processUser.jsp page to process the record.

```
7 <%@page import="java.sql.*"%>
8 <%@page contentType="text/html" pageEncoding="UTF-8"%>
9 <!DOCTYPE html>
10 <html>
11 <head>
12 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
13 <title>JSP Page</title>
14 </head>
15 <body>
16 <%
17     String username = request.getParameter("Username");
18     String password = request.getParameter("Password");
19     String firstName = request.getParameter("FirstName");
20     String lastName = request.getParameter("LastName");
21
22     Class.forName("com.mysql.jdbc.Driver");
23
24     String query = "INSERT INTO userprofile (username, password, firstname, lastname) VALUES (?, ?, ?, ?)";
25
26     try (Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/cse3023", "root", "admin");
27         PreparedStatement pstmt = con.prepareStatement(query)) {
28
29         pstmt.setString(1, username);
30         pstmt.setString(2, password);
31         pstmt.setString(3, firstName);
32         pstmt.setString(4, lastName);
33
34         pstmt.executeUpdate();
35     } catch (SQLException e) {
36         e.printStackTrace();
37         System.out.println("Error message" + e);
38     }
39 <%
40
41 <div>
42 <h1>Processing User Information</h1>
43
44 <p>Username: <%= username%></p>
45 <p>Password: <%= password%></p>
46 <p>First Name: <%= firstName%></p>
47 <p>Last Name: <%= lastName%></p>
48
49 <button><a href="login.jsp">Go to Login</a></button>
50 </div>
51 </body>
52 </html>
53
```

Processing User Information

Username: caiyi

Password: 123

First Name: cai yi

Last Name: wong

[Go to Login](#)

4. Create login.jsp page to login to the system.

```
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <!DOCTYPE html>
9 <html>
10 <head>
11 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
12 <title>JSP Page</title>
13 </head>
14 <body>
15 <div>
16 <form action="doLogin.jsp" method="POST">
17 <h1>User Login</h1>
18
19 <label>Username: </label>
20 <input type="text" name="Username" id="username" required/>
21
22 <label>Password: </label>
23 <input type="password" name="Password" id="password" required/>
24
25 <p>
26 <input type="submit" value="User Login" id="btn"/>
27 </p>
28 </form>
29 </div>
30 </body>
31 </html>
```

User Login

Username: Password:

5. Create doLogin.jsp to validate username and password. If validation is successful, redirect the page to main.jsp page that displays the username, firstname and lastname.

```
7  <%@page import="java.sql.*"%>
8  <%@page contentType="text/html" pageEncoding="UTF-8"%>
9  <!DOCTYPE html>
10 <html>
11 <head>
12 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
13 <title>JSP Page</title>
14 </head>
15 <body>
16 <%
17     String username = request.getParameter("Username");
18     String password = request.getParameter("Password");
19
20     Class.forName("com.mysql.jdbc.Driver");
21
22     String query = "SELECT * FROM userprofile WHERE username = ? AND password = ?";
23     ResultSet rs = null;
24
25     try (Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/cse3023", "root", "admin");
26         PreparedStatement pstmt = con.prepareStatement(query)) {
27
28         pstmt.setString(1, username);
29         pstmt.setString(2, password);
30
31         rs = pstmt.executeQuery();
32
33         if(rs.next()){
34             session.setAttribute("username", username);
35             session.setAttribute("firstName", rs.getString("firstName"));
36             session.setAttribute("lastName", rs.getString("lastName"));
37             response.sendRedirect("main.jsp");
38         } else {
39             out.println("<p style='color:red'>Invalid username or password!</p>");
40             request.getRequestDispatcher("login.jsp").include(request, response);
41         }
42     } catch (SQLException e) {
43         e.printStackTrace();
44         System.out.println("Error message" + e);
45     }
46 <%>
47 </body>
48 </html>
49
50
```

Welcome, caiyi !

First Name: cai yi

Last Name: wong

If incorrect, will display an error message

User Login

Username: Password:

User Login

Invalid username or password!